

# DisruptIQ V2

The autonomous disruption-response layer for supply chains.

Nine specialised AI agents detect a disruption, recall similar past incidents, forecast demand impact, score per-supplier risk, and deliver three ranked, Monte-Carlo-simulated recovery actions - in under 90 seconds, with mandatory human sign-off.

**< 90s**

full swarm runtime

**9**

coordinated agents

**3**

novel algorithms (MCF / CCS / MSDS)

**100%**

decisions human-gated

Theme: Agentic AI - Responsible AI by Design

## THE PROBLEM

# Supply chains see risk. They can't respond to it.

# \$184M

average annual loss per enterprise from supply-chain disruption.

# 55%

of supply-chain leaders have no visibility into their upstream network - the blind spot where the cost lives.

## The market's structural gap

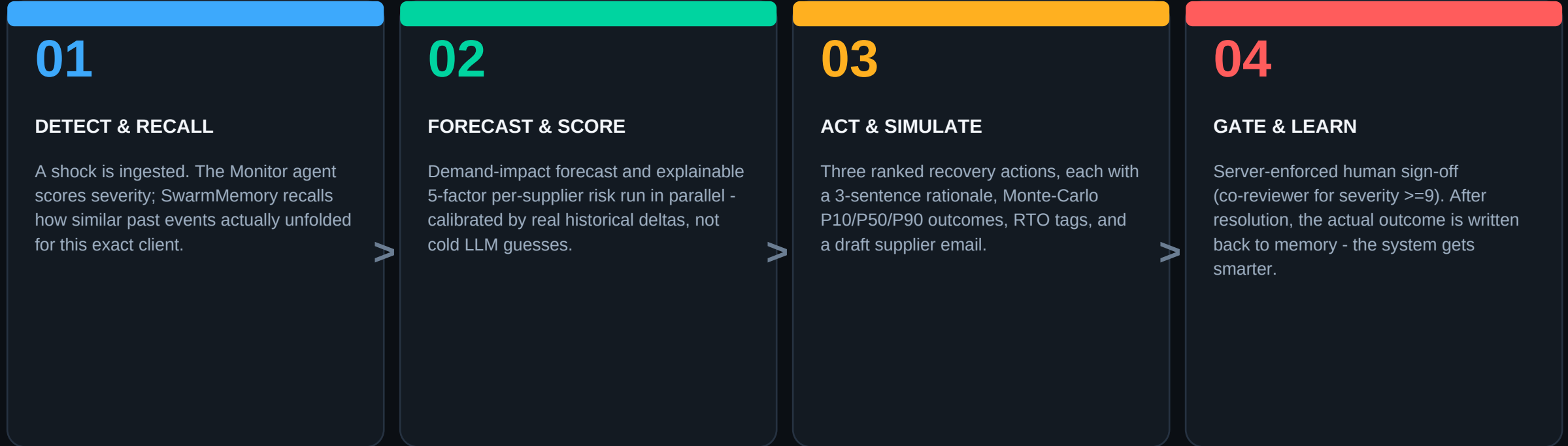
- **Incumbents detect.**  
Resilinc, Everstream, Prewave aggregate billions of signals and raise alarms.
- **Humans still execute.**  
Evaluating alternates, forecasting demand, drafting outreach - all manual, under pressure.
- **Alert fatigue wins.**  
Dashboards flood teams with probabilities; nobody is told what to actually do.

## Why now

Just-in-time inventory has been gutted by cyclones, port strikes, supplier insolvency, tariff shocks, and the weaponisation of critical materials. Risk management has moved from an operational chore to a board-level mandate. The market no longer wants dashboards that merely highlight risk - it wants a system that ingests the shock, simulates the financial impact, and prescribes the optimal mitigation. That is an execution problem, and execution is exactly where today's intelligence-aggregator platforms stop.

# From visibility to autonomous orchestration.

DisruptIQ is not another risk dashboard. It is the execution layer that turns a disruption into a decision.



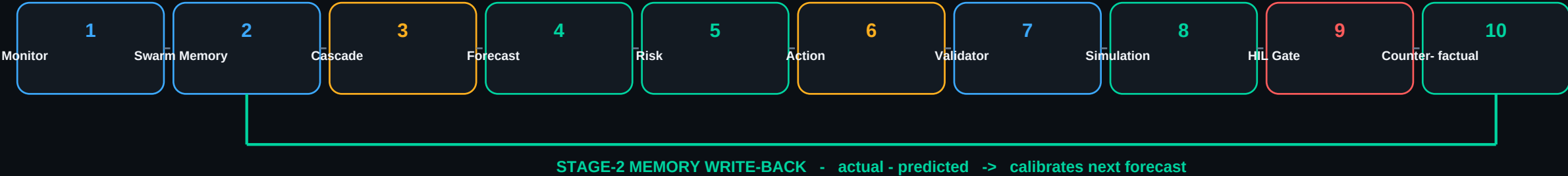
## The pitch in one line

DisruptIQ delivers the speed of a 9-agent swarm with the auditability enterprise compliance demands - resolution over detection, learning over alerting.

# MCAS - the Memory-Calibrated Agent Swarm.

A stateless swarm that becomes stateful: counterfactual write-backs calibrate every future forecast.

THE 9-AGENT PIPELINE (<= 90s SLA)



MCF

Memory-Calibrated Forecast

forecast = base + w \* mean(actual - predicted).  
Bayesian-style online calibration driven by counterfactual write-backs.

backend/algorithms.py

CCS

Compound Cascade Severity

combined = max(sa, sb) \* mult \* (1 + shared/total).  
Escalates severity by real supplier-overlap ratio across compound events.

backend/algorithms.py

MSDS

Multi-Signal Dissent Score

divergence = |forecast - risk| / max. If > 0.15, fire a server-enforced human checkpoint. Kills silent hallucination.

backend/algorithms.py

# Responsible AI by design - built for Microsoft Build.

Generative reasoning where it adds value; deterministic guardrails everywhere it matters.

## GUARDRAILS

- **Human-in-the-loop is server-enforced**  
The API rejects any action confirmation missing the required acknowledgments. The UI cannot bypass it.
- **Co-reviewer for severity  $\geq 9$**   
Two humans must approve the most critical decisions before the swarm proceeds.
- **Azure Content Safety**  
Every AI narrative is filtered before it is ever displayed to an operator.
- **Learning is human-gated**  
Stage-2 memory is only written after a human confirms the actual outcome. Humans decide what the system learns.

## EXPLAINABILITY & RESILIENCE

- **Explainable risk scoring**  
Every supplier risk exposes a 5-factor breakdown (proximity, buffer, sites, reliability, category) via a 'Why?' drill-down.
- **Graceful degradation**  
When LLM quota is exhausted, agents fall back to dataset-aware deterministic logic - never hallucinate, never fail silently.
- **Hallucination guard**  
The Action agent maps every generated supplier ID back to the client's real database; invented IDs are replaced.
- **Full audit trail**  
Every agent decision and human action is logged with `client_id`, `timestamp`, and `actor`.

Stack: [GitHub Models \(LLM\)](#) - [Azure Content Safety](#) - [Azure Cosmos DB](#) - [FastAPI](#) + [Socket.IO](#) - [React 18](#) + [Vite](#)

# Everyone detects. We orchestrate.



## Our uncontested whitespace

Incumbents sell visibility and alerting - built to aggregate external data and sound the alarm. The operational burden of resolving the disruption falls back on humans. DisruptIQ owns that gap: the Disruption-Response Orchestrator with epistemic memory.

- **Learning organisation**  
MCF write-backs make the platform a compounding, client-bespoke asset - not a generic public-data model.
- **Deterministic Safe-AI**  
Sub-90s swarm + server-enforced gates + explainable scoring: speed with auditability incumbents lack.
- **Resolution, not alert fatigue**  
Three ranked, simulated actions with draft emails - not thousands of probabilistic alerts.

# Where DisruptIQ wins, and where we're building.

| Capability                   | Resilinc | Everstream | Prewave | Elementum | DisruptIQ |
|------------------------------|----------|------------|---------|-----------|-----------|
| Autonomous orchestration     | ~        | -          | ~       | Y         | Y         |
| Counterfactual memory loop   | -        | -          | -       | -         | Y         |
| Server-enforced HIL gates    | ~        | -          | -       | Y         | Y         |
| Explainable 5-factor risk    | ~        | ~          | -       | -         | Y         |
| Sub-90s resolution actions   | -        | -          | -       | ~         | Y         |
| Proactive auto-monitoring    | Y        | Y          | Y       | -         | >         |
| N-tier visibility            | Y        | Y          | Y       | -         | >         |
| Native ERP / TMS integration | Y        | Y          | ~       | Y         | >         |
| Financial / ESG signals      | ~        | ~          | Y       | -         | >         |
| Global geographic coverage   | Y        | Y          | Y       | Y         | >         |

Y Strong / shipped      ~ Partial      - Absent      > On roadmap

THE PRODUCT

# Live, multi-tenant, and deployed.

Strict client\_id isolation: a real client never sees demo data or another tenant's data.

SWARM FEED

Monitor

severity 8.4

Forecast

-22% demand

Risk

3 critical

Action

3 options ready

HIL Gate

awaiting sign-off

SUPPLIER RISK

Apex Components

CRITICAL 82

Meridian Steel

HIGH 67

Coastal Logistics

HIGH 61

Inland Parts

MEDIUM 48

Why? 5-factor drill-down

explainable

RECOVERY ACTION #1

Re-route via alt port

P50 +4 days

Cost premium

+9.2%

Monte Carlo P10/P90

+2 / +8 days

Draft supplier email

ready

RTO tag

moderate

What's shipped

- 9-agent swarm + 4 enhancement agents
- Owner-only Admin Console (10 tabs)
- 10 industry sample datasets (270+ suppliers)
- CSV / Excel onboarding + export
- 100+ REST endpoints, real-time Socket.IO
- Premium tiering + upgrade flow
- Resilience score, dependency heatmap, Tier-2 view
- JWT auth, sessions, rate-limiting, Content Safety

Deployed: Railway (backend + frontend) - [disruptiqai.com](https://disruptiqai.com)



# Closing the ingestion gap to go enterprise-grade.

Architecture is our moat. These build priorities turn a tier-one prototype into a tier-one platform.

01

CRITICAL

Proactive auto-monitoring API

Cross-reference live news / NOAA / disaster feeds against supplier coordinates; auto-fire the Monitor agent on threshold breach.

HIGH feasibility

02

CRITICAL

Native ERP / TMS integration

Bidirectional connectors (SAP, Oracle NetSuite, major TMS) to ingest live buffer days, lead times, in-transit status.

MODERATE feasibility

03

HIGH

Financial & compliance signals

D&B / OpenCorporates / sanctions ingestion; add Financial Viability + ESG weights to the Risk agent.

HIGH feasibility

04

HIGH

Cross-client federated learning

Anonymise counterfactual deltas into a Global Baseline Memory - solves the cold-start problem on day one.

MODERATE feasibility

05

MEDIUM

Probabilistic Tier-N graphing

Tier-2 view shipped; extend via BOM NLP or a knowledge-graph partner to surface deep single-source chokepoints.

BUILDING

# The execution layer for modern supply-chain risk.

- **Novel architecture**  
MCAS + 3 named algorithms (MCF / CCS / MSDS) - a learning loop incumbents structurally lack.
- **Responsible AI by design**  
Server-enforced human gates, Content Safety, explainability, graceful degradation.
- **Real, deployed, multi-tenant**  
100+ endpoints, strict client isolation, live on Railway under disruptiqai.com.

TEAM

## DisruptIQ

Built for the Microsoft Build AI Hackathon 2026 under the Agentic AI theme.

|       |                         |
|-------|-------------------------|
| Stack | FastAPI - React - Azure |
| LLM   | GitHub Models           |
| Data  | Cosmos DB               |
| Live  | disruptiqai.com         |

Team size: 1-3 - Theme: Agentic AI